

CSCI 240 PA1 Submission

Exercise 1

Source code below:

```
class Ex1 {
    public static int countDuplicates(int[] arr) {

        int count = 0;

        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    count++;
                }
            }
        }
        return count;
    }

    public static void main(String[] args) {

        System.out.println("Author: Aiden Wang\n");

        int[] test1 = {5, 7, 2, 6, 4};
        int[] test2 = {5, 7, 2, 2, 5, 6, 4, 2};

        System.out.println("Test 1 (5, 7, 2, 6, 4) result: " +
countDuplicates(test1));
        System.out.println("Test 2 (5, 7, 2, 2, 5, 6, 4, 2) result: " +
countDuplicates(test2));
    }
}
```

Input/output below:

```
/Users/guoqingwang/Library/Java/JavaVirtualMachines
Author: Aiden Wang

Test 1 (5, 7, 2, 6, 4) result: 0
Test 2 (5, 7, 2, 2, 5, 6, 4, 2) result: 4

Process finished with exit code 0
```

Exercise 2

Source code below:

```
import static PA1.Ex1.countDuplicates;

import java.util.Random;

public class Ex2 {

    public static void main(String[] args) {

        System.out.println("Author: Aiden Wang\n");
        Random rand = new Random();
        System.out.println("| People | Count | Percent |");
        System.out.println("| ----- | ----- | ----- |");

        for (int people = 5; people <= 100; people += 5) {
            int sharedBirthdayCount = 0;

            for (int exp = 0; exp < 100; exp++) {
                int[] birthdays = new int[people];
                for (int i = 0; i < people; i++) {
                    birthdays[i] = rand.nextInt(365) + 1;
                }
                if (countDuplicates(birthdays) > 0) {
                    sharedBirthdayCount++;
                }
            }
            System.out.printf("| %-6d | %-5d | %d%%\n", people,
sharedBirthdayCount, sharedBirthdayCount);
        }
    }
}
```

Input/output below:

```
/Users/guoqingwang/Library/Java/JavaVirtualMachines/temurin-22.0.2/Contents/Home/bin/java -javaagent:/Applications
Author: Aiden Wang

| People | Count | Percent |
| ----- | ----- | ----- |
| 5       | 3      | 3%      |
| 10      | 13     | 13%     |
| 15      | 23     | 23%     |
| 20      | 41     | 41%     |
| 25      | 61     | 61%     |
| 30      | 66     | 66%     |
| 35      | 80     | 80%     |
| 40      | 90     | 90%     |
| 45      | 94     | 94%     |
| 50      | 100    | 100%    |
| 55      | 97     | 97%     |
| 60      | 99     | 99%     |
| 65      | 100    | 100%    |
| 70      | 100    | 100%    |
| 75      | 100    | 100%    |
| 80      | 100    | 100%    |
| 85      | 100    | 100%    |
| 90      | 100    | 100%    |
| 95      | 100    | 100%    |
| 100     | 100    | 100%    |

Process finished with exit code 0
```

Exercise 3

Source code below:

```
package PA1;

public class Ex3 {

    public static class DiffProgression extends Progression {
        protected long prev;

        public DiffProgression() {
            this(2, 200);
        }

        // Two-argument constructor
        public DiffProgression(long first, long second) {
            super(first);
            prev = first - second;
        }
        @Override
        protected void advance() {
            long temp = prev;
            prev = current;
            current = Math.abs(current - temp)
;
        }
        public static void main(String[] args) {
            System.out.println("Author: Aiden Wang\n");
            System.out.print("Default DiffProgression: ");
            Progression p1 = new DiffProgression();
            p1.printProgression(5);
            System.out.print("DiffProgression with 15 and 3: ");
            Progression p2 = new DiffProgression(15, 3);
            p2.printProgression(5);
        }
    }
}
```

Input/output below:

```
/Users/guoqingwang/Library/Java/JavaVirtualMachines/temurin
Author: Aiden Wang

Default DiffProgression: 2 200 198 2 196
DiffProgression with 15 and 3: 15 3 12 9 3

Process finished with exit code 0
```

Exercise 4

Source code below:

```
package PA1;

import java.util.ArrayList;
import java.util.List;

public class Ex4 {
    public static class MyCreditCard extends CreditCard {
        private final double apr;
        private final List<String> transactions;

        public MyCreditCard(String cust, String bk, String acnt, int lim,
double initialBal, double apr) {
            super(cust, bk, acnt, lim, initialBal);
            this.apr = apr;
            this.transactions = new ArrayList<>();
        }

        @Override
        public boolean charge(double price) {
            boolean isSuccess = super.charge(price);
            if (isSuccess) {
                transactions.add("Charge: $" + String.format("%.2f",
price));
            }
            return isSuccess;
        }

        @Override
        public void makePayment(double amount) {
            super.makePayment(amount);
            transactions.add("Payment: $" + String.format("%.2f",
amount));
        }

        public void addInterest() {
            if (balance > 0) {
                double monthlyFactor = Math.pow(1 + apr, 1.0 / 12) - 1;
                double interest = balance * monthlyFactor;
                balance += interest;
                transactions.add("Interest: $" + String.format("%.2f",
interest));
            }
        }

        public void printTransactions() {
            System.out.println("Acct " + getAccount() + " History:");
            System.out.print("  ");
            for (int i = 0; i < transactions.size(); i++) {
                System.out.print(transactions.get(i));
                if (i < transactions.size() - 1) {
                    System.out.print(" | ");
                }
            }
        }
    }
}
```

```
        if ((i + 1) % 4 == 0 && i != transactions.size() - 1) {
            System.out.print("\n    ");
        }
    }
    System.out.println("]");
}

public static void main(String[] args) {
    System.out.println("Author: Aiden Wang\n");
    // 1. Create 2 credit cards with annual interest of 10%
    MyCreditCard card1 = new MyCreditCard("Aiden", "Bank of Java",
"1234", 5000, 0, 0.10);
    MyCreditCard card2 = new MyCreditCard("Alice", "C++
Federation", "5678", 2000, 0, 0.10);

    System.out.println("--- 6-Month Simulation ---");
    // 2. Simulate 6 months
    for (int month = 1; month <= 6; month++) {
        // Card 1 operations
        card1.charge(300.0);
        card1.charge(50.0);
        card1.makePayment(100.0); // Partial payment
        card1.addInterest();

        // Card 2 operations
        card2.charge(150.0);
        card2.makePayment(150.0); // Full payment
        card2.addInterest();

        // Print both balances compactly on a single line
        System.out.printf("Month %d | Card 1 Bal: $%-7.2f | Card 2
Bal: $%.2f\n",
            month, card1.getBalance(), card2.getBalance());
    }

    // 3. Print all transactions
    System.out.println("\n--- Transaction History ---");
    card1.printTransactions();
    card2.printTransactions();
}
}
```

Due Date: 3/5/2026

Names: Aiden Wang

Input/output below:

```
/Users/guoqingwang/Library/Java/JavaVirtualMachines/temurin-22.0.2/Contents/Home/bin/java -javaagent:/Application
Author: Aiden Wang

--- 6-Month Simulation ---
Month 1 | Card 1 Bal: $251.99 | Card 2 Bal: $0.00
Month 2 | Card 1 Bal: $506.00 | Card 2 Bal: $0.00
Month 3 | Card 1 Bal: $762.02 | Card 2 Bal: $0.00
Month 4 | Card 1 Bal: $1020.09 | Card 2 Bal: $0.00
Month 5 | Card 1 Bal: $1280.22 | Card 2 Bal: $0.00
Month 6 | Card 1 Bal: $1542.43 | Card 2 Bal: $0.00

--- Transaction History ---
Acct 1234 History:
[Charge: $300.00 | Charge: $50.00 | Payment: $100.00 | Interest: $1.99 |
Charge: $300.00 | Charge: $50.00 | Payment: $100.00 | Interest: $4.00 |
Charge: $300.00 | Charge: $50.00 | Payment: $100.00 | Interest: $6.03 |
Charge: $300.00 | Charge: $50.00 | Payment: $100.00 | Interest: $8.07 |
Charge: $300.00 | Charge: $50.00 | Payment: $100.00 | Interest: $10.13 |
Charge: $300.00 | Charge: $50.00 | Payment: $100.00 | Interest: $12.20]
Acct 5678 History:
[Charge: $150.00 | Payment: $150.00 | Charge: $150.00 | Payment: $150.00 |
Charge: $150.00 | Payment: $150.00 | Charge: $150.00 | Payment: $150.00 |
Charge: $150.00 | Payment: $150.00 | Charge: $150.00 | Payment: $150.00]

Process finished with exit code 0
```

Answer for Question 1

List and describe some typical components in a class:

A class is a blueprint that encapsulates data and behavior into a single unit.

Core Components of a Class

Fields (Attributes): The State. Variables that store data.

Methods (Behaviors): The Logic. Functions that define what the object does.

Constructors: The Initialization. Special blocks that set up the object when it was created.

Access Modifiers: The Visibility. Keywords (public, private) that enforce Encapsulation.

The "this" Reference: The Identity. A pointer to the specific instance currently being executed.

The "3-Layer" Essence

Data: What it knows (Fields).

Action: What it does (Methods).

Contract: How it interacts (Constructors & Modifiers).

Answer for Question 2

What are some good reasons for taking advantage of inheritance instead of creating a new standalone class?

Inheritance allows a "Subclass" to derive properties from a "Superclass," creating an "is-a" relationship.

The "4-C" Advantage

Code Reuse : Inherit existing logic automatically. Don't reinvent the wheel; just add the spokes.

Consistency: Ensure all related classes follow the same Interface. If the parent has print(), every child is guaranteed to have print().

Change Management: Fix a bug in the Parent Class, and it instantly repairs all Subclasses.

Core Polymorphism: Treat different objects (e.g., Square, Circle) as one type (Shape). This allows for flexible, scalable code.

The Bottom Line

Standalone Class: A "Silo." You build everything from scratch.

Inheritance: An "Evolution." You take what works and only code the Differences.

Extra Credit Option 1

Source code below:

```
package PA1;

import java.util.Random;

public class EC1 {

    public static void main(String[] args) {
        System.out.println("Author: Aiden Wang\n");
        String baseSentence = "I will never spam my friends again.";
        int numLines = 50;
        int numMistakes = 10;
        String[] sentences = new String[numLines];
        int[] mistakeCounts = new int[numLines];
        for (int i = 0; i < numLines; i++) {
            sentences[i] = baseSentence;
        }
        Random rand = new Random();
        // 1. Distribute exactly 10 mistakes randomly across the 50 lines
        for (int i = 0; i < numMistakes; i++) {
            mistakeCounts[rand.nextInt(numLines)]++;
        }
        // 2. Apply the mistakes
        for (int i = 0; i < numLines; i++) {
            if (mistakeCounts[i] > 0) {
                char[] chars = sentences[i].toCharArray();
                for (int m = 0; m < mistakeCounts[i]; m++) {
                    int charIdx;

                    do {
                        charIdx = rand.nextInt(chars.length);
                    } while (!Character.isLetter(chars[charIdx]));

                    char oldChar = chars[charIdx];
                    char newChar;
                    do {
                        newChar = (char) ('a' + rand.nextInt(26));
                    } while (newChar == Character.toLowerCase(oldChar));

                    if (Character.isUpperCase(oldChar)) {
                        newChar = Character.toUpperCase(newChar);
                    }

                    chars[charIdx] = newChar;
                }

                sentences[i] = new String(chars) + " (" + mistakeCounts[i]
+ " mistakes)";
            }

            System.out.printf("%2d: %-52s", i + 1, sentences[i]);

            if ((i + 1) % 2 == 0) {
                System.out.println();
            }
        }
    }
}
```


Due Date: 3/5/2026

Names: Aiden Wang

```
    } else {  
        System.out.print(" | ");  
    }  
}  
}  
}
```

Input/output below:

1st run:

```
/Users/guoqingwang/Library/Java/JavaVirtualMachines/temurin-22.0.2/Contents/Home/bin/java -javaagent:/Applications/IntelliJ IDEA.app  
Author: Aiden Wang  
  
1: I will never spam my friends again. | 2: I will never spam my friends again.  
3: I will never spam my friends again. | 4: I will never spam my friends again.  
5: I will never spam my friends again. | 6: I will never spam my friends again.  
7: I will never spam my friends again. | 8: I will never spam my frilnds again. (2 mistakes)  
9: I will never spam my friends again. | 10: I will never spam my friends again.  
11: I will never spam my friends again. | 12: I will never spam my friends again.  
13: I will never spam my friends again. | 14: I will never spam my friends again.  
15: I will never spam my friends again. | 16: I will never spam my friends again.  
17: I will never spam my friends again. | 18: I will never spam my friends again.  
19: I will never spam my friends again. | 20: I will never spam my friends again.  
21: I will never spam my friends again. | 22: I will never spam my friends again.  
23: I will never spam my friends again. | 24: I will never spam my friends again.  
25: I will never slam my friends again. (1 mistakes) | 26: I will never spam my friends again.  
27: I will never spam my friends again. | 28: I will never spam my friends again.  
29: I will never spam my friends again. | 30: I will never spam my friends again.  
31: I will nvver spam my friends again. (1 mistakes) | 32: I will never ppam my friends again. (1 mistakes)  
33: I will never spam my friends again. | 34: I will never spam my friends again.  
35: I will zever spam my friends again. (1 mistakes) | 36: I will never spam my friends again.  
37: I will never spam my friends again. | 38: I will never spam my friends again.  
39: I will never spaz my friends again. (1 mistakes) | 40: I will never spam my friends again.  
41: I will never spam my friends again. | 42: I will never spam my friends again.  
43: I will never spam my friends again. | 44: I wilh never spam my friends again. (1 mistakes)  
45: I will never spam my friends again. | 46: I will never spam my friends again.  
47: I will never spam my friends again. | 48: W will never spam my friends again. (1 mistakes)  
49: I will never spam gy friends again. (1 mistakes) | 50: I will never spam my friends again.  
  
Process finished with exit code 0
```

2nd run:

```
/Users/guoqingwang/Library/Java/JavaVirtualMachines/temurin-22.0.2/Contents/Home/bin/java -javaagent:/Applications/IntelliJ IDEA.app/Cor  
Author: Aiden Wang  
  
1: I will never spam yy friends again. (1 mistakes) | 2: I will never spam my friends again.  
3: I will never spam my friends again. | 4: I will never spam my friends again.  
5: I will never spam ay friends again. (1 mistakes) | 6: I will never spam my friends again.  
7: I will never spam my friends again. | 8: I will never spam my friends again.  
9: I will never spam my friends again. | 10: I will never spam my friends again.  
11: I will never spam my friends again. | 12: I will nevwr spam my friends aaain. (2 mistakes)  
13: I will never spam my friends again. | 14: I will never spam my friends again.  
15: I will never spam my friends again. | 16: I wslr never spam my friends again. (1 mistakes)  
17: I will never spam my friends again. | 18: I will never spam my friends again.  
19: I will never spam my friends again. | 20: I will never spam my friends again.  
21: I will never spam my friends again. | 22: I will never spam my friends again.  
23: I will never spam my friends again. | 24: I will never spam my friends again.  
25: I will never spam my friends again. | 26: I zill never spam my frijnds again. (2 mistakes)  
27: I will never spam my friends again. | 28: I will never spam my friends again.  
29: I will never spam my friends again. | 30: I will never spam my friends again.  
31: I will never spam my friends again. | 32: I will never spam my friends again.  
33: I will never spam my friends again. | 34: I will never spam my friends again.  
35: I will never spam my friends again. | 36: I will never spam my friends again.  
37: I will never spam my friends again. | 38: I will never spam my friends again.  
39: I will never spam my friends again. | 40: I will never spam my friends again.  
41: I will never spam my friends again. | 42: I will never spam my friends again.  
43: I will never spam my friends again. | 44: I will never spam my friends again.  
45: I will never spam my friends again. | 46: I will never spam my friends agein. (1 mistakes)  
47: I will gevery spm my friends again. (2 mistakes) | 48: I will never spam my friends again.  
49: I will never spam my friends again. | 50: I will never spam my friends again.  
  
Process finished with exit code 0
```